Ken Wu

(647)-918-1903 | kj3wu@uwaterloo.ca | LinkedIn | Github

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science and Finance — *Sep 2023 – May 2028*

- **President's Scholarship of Distinction**, University of Waterloo
- **1st Place, Most Profitable**, Waterloo Derivatives and Securities Paper Trading Competition — Apr 2026
- **3rd Place**, Waterloo Algorithmic Trading Competition — Dec 2024

Experience

RECO — Toronto, ON
Software Developer — *May 2026 – Aug 2026*

- Engineered a **FastAPI** compliance platform on Azure that **screens up to 4,000 real estate ads weekly and cuts manual review time by 50%** across the compliance team.
- Integrated **Azure Computer Vision** to analyze advertisement images for regulated content, automatically detecting violations such as missing brokerage information, misleading visuals, and unauthorized signage.
- Leveraged **Azure OpenAI** to parse advertisement text and generate compliance **flags** against regulatory rules, **reducing over two hrs/day** of manual labour for officers.
- Implemented a **semantic caching layer** for the internal LLM service: responses from an initial batch of queries are stored, and new queries are matched by **similarity search** against the cache to return stored outputs, reducing redundant LLM calls and latency.
- Introduced an **Azure SQL** database schema for advertisement metadata, **increasing DB reads by 50%**.
- **Cut false positives by 25%** by building an **evaluation harness** that replays labeled ads to score precision and recall.
- Demoed the platform to 200+ industry professionals, including the Minister, at the **Collaboration Summit** in Ottawa.

Geotab — Toronto, ON
AI Platform Developer — *Jan 2025 – May 2025*

- Automated internal workflows by building **generative AI agents**, **saving developers 5 hours weekly** on manual tasks.
- Saved Geotab **over $1,700 daily** by building a cost-saving agent with **self-evaluation** and **reinforcement learning** on Vertex AI and Gemini 2.5 Pro to optimize Google Cloud CPU usage.
- Improved retrieval accuracy and response times by **30%** by re-engineering the Superset **dashboard management** agent with **metadata-indexed RAG** from GCS buckets.

Software Developer — *May 2025 – Aug 2025*

- Led a refactor of the **GenAI Gateway**'s core modules by decoupling provider integrations behind a common interface, improving **latency by 25%** and **downtime by 30%**, and making it easier to onboard new models and providers.
- Reduced the time to onboard a new AI model from 3 days to 2 hours by building an automated **load-testing pipeline** in **GitLab CI** using **Locust**, replacing manual performance checks.
- **Prevented 60+ slow model releases** by adding automated performance checks to the Locust CI pipeline that block a merge request if a new AI model's **p95 latency** or error rate regresses.

Microsoft - Azure & Artificial Intelligence — Remote (Waterloo, ON)
Project Team Member — *May 2024 – Aug 2024*

- Created a working prototype of a financial investment coach, targeting a projected 2% monthly return using **Azure AI** and Azure services, including Azure Bot Services and Azure App Services.
- Collaborated with industry professionals to deliver a **production-level** product, combining strategy and cross-team coordination.
- Validated the projected **2% monthly return** by building a **backtesting pipeline** with **pandas** and **NumPy** that replays the strategy on historical data to gate bot output.

Projects

Coinbase Algorithmic Trader | *Python, MongoDB, REST, Docker, Pandas, Prophet*

- Generated **0.5% average weekly returns** across **1,000+ daily transactions** with an algorithmic trading platform built on **Python**, **Coinbase REST APIs**, and quantitative analysis models.
- Integrated and deployed a trading advisor bot with the **Telegram API** and **Streamlit Cloud** using **Docker** and CI/CD, improving accessibility and real-time strategy monitoring.

Agentic Playlist Generator | *Python, Flask, Spotify Web API, OAuth2, RAG, JavaScript*

- Built an **agentic playlist generator** on the **Spotify Web API** and **OAuth2** that matches a user's mood or activity prompt against their listening history and track metadata.
- Implemented an **agent** with a **retrieval and reranking** stage to select and rank candidate tracks against the prompt, plus a **tracer** that logs each agent step to debug and compare runs.
- **Improved relevance by 50%** over the baseline by building an **evaluation pipeline** that scores playlists on relevance and diversity to compare retrieval and reranking changes.

Technical Skills

Languages: Python, TypeScript, C++, C, Kotlin, Java, JavaScript, HTML/CSS, Dart
Developer Tools: Git, Docker, Kubernetes, Helm, Terraform, GitLab CI/CD, AWS, Azure, Google Cloud Platform
Technologies: FastAPI, REST APIs, Azure SQL, MongoDB, BigQuery, RAG, OpenAI API, Vertex AI, pandas, NumPy